

Course Code: 503-01**Course Title: Advance Web Designing**

Course Code	503-01									
Course Title	Advance Web Designing (Major-11-01)									
Credits	4									
Course Category	Major Course									
Level of Course	400-499 (Advance Level)									
Teaching per week	2 Hours Theory + 4 Hours of Lab/interactive Project work.									
Minimum Hours per Semester	90 Hours (30 Hours Theory + 60 Hours of Project work (Including class work, examination, preparation etc.)									
Review / Revision	-									
Implementation Year:	A.Y. 2026-2027									
Cognitive Skills of the Course	Understand the technical foundations, as well as the non-programming / administrative skills needed to be a successful web developer. This course reveals the reasons why a truly successful website developer does more than write code. The course deals with both the Frontend (client-side) and Backend (server-side) of a tech product. This course deals with designing of websites and building the Applications.									
Course Objective	The students will learn the whole React WebApp building process, from pc to the server. They will work with NoSQL databases. They will learn the whole process of building your App using React.js. At the end of the course, they will develop modern, complex, responsive and scalable web applications with Angular.									
Pre-requisite	Paper-305-01 (Web Designing -I) in Semester-3. Paper-405-01 (Web Designing -II -2) in Semester-4.									
Course Outcomes	CO1 (Understanding): Understand core concepts of web development and their applications in modern digital environments. CO2 (Understanding & Applying): Identify career relevance of web technologies in development, design, and entrepreneurship. CO3 (Applying): Apply knowledge through hands-on exercises to build interactive and functional websites. CO4 (Analyzing): Analyze user and system requirements to plan and structure effective web solutions. CO5 (Creating): Develop responsive, scalable, and modern websites using appropriate design and coding practices. CO6 (Understanding & Applying): Understand and implement key client-side and server-side components in web development.									
Mapping between Course Outcomes(CO) with Program Specific Outcomes(PSO)			PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PSO7	PSO8
	CO1									
	CO2									
	CO3									
	CO4									
	CO5									
	CO6									
Course Content	Unit-1: Concepts of NoSQL: MongoDB 1.1 concepts of NoSQL. Advantages and features. 1.1.1 MongoDB Datatypes (String, Integer, Boolean, Double, Arrays, Objects) 1.1.2 Database creation and dropping database 1.2 create and Drop collections 1.3 CRUD operations (Insert, update, delete, find, Query and Projection operators) 1.4 Operators (Projection, update, limit(), sort()) and Aggregation commands Unit-2: Fundamentals of React.js 2.1 Overview of React 2.1.1 Concepts of React. 2.1.2 Using React with HTML 2.1.3 React Interactive components: Components within components and Files 2.1.3 Passing data through Props 2.2 Class components									

	2.2.1 React class and class components 2.2.2 Conditional statements, Operators, Lists 2.2.3 React Events: Adding events, Passing arguments, Event objects 2.3 Forms: (Adding forms, Handling forms, Submitting forms) 2.3.1 event.target.name and event.Target.event, React Memo 2.3.2 Components (TextArea, Drop down list (SELECT)) 2.4 Hooks: Concepts and Advantages 2.4.1 useState, useEffect, useContext 2.4.2 useRef, useReducer, useCallback, useMemo 2.4.3 Hook: Building custom hook, advantages and use Unit-3: Fundamentals of Angular 3.1 Concepts and Characteristics of Angular 3.1.1 Key Concepts 3.1.1.1 TypeScript Overview, Component-based Architecture 3.1.1.2 Modules, Services, Dependency Injection 3.2 Routing and Navigation3 3.2.1 Setting up Development Environment, Installing Node.js and Angular CLI, Creating and Running Angular Projects, Project Structure and Angular Workspace Unit-4: Angular Features 4.1 Angular Core Features 4.1.1 Angular Components, Creating Components (@Component) 4.1.2 Component Lifecycle Hooks 4.2 Component Communication (Input, Output, EventEmitter) 4.2.1 Angular Directives, Structural Directives: *ngIf, *ngFor, *ngSwitch 4.2.2 Attribute Directives: ngClass, ngStyle 4.3 Custom Directives 4.3.1 Angular Data Binding 4.3.2 Interpolation, Property Binding, Event Binding, Two-way Binding with [(ngModel)] 4.4 Angular Pipes (Filters in Angular) 4.4.1 Built-in Pipes: uppercase, lowercase, currency, date, percent, titlecase, slice, Creating Custom Pipes 4.5 Angular Forms (Template-driven Forms, Reactive Forms, Form Validation) 4.6 Angular Services and Dependency Injection 4.6.1 Creating and Injecting Services, Using HttpClient to call REST APIs 4.6.2 Observables with RxJS
Reference Books	1. Web Development with Node and Express, Ethan Brown, O'Reilly Media, Inc., ISBN: 978-1-491-94930-6 2. Node.js, MongoDB, React, React Native Full-Stack Fundamentals and Beyond, Eric Bush, Blue Sky Productions Inc., ISBN: 978-0-9971966-8-9 3. MongoDB Fundamentals: A hands-on guide to using MongoDB and Atlas in the real world, Amit Phaltankar, Juned Ahsan, Michael Harrison, Liviu Nedov , ISBN:978-1-83921-064-8 4. Sams Teach Yourself NoSQL with MongoDB in 24 Hours, Pearson Education ISBN-13: 9780672337130 5. MongoDB Basics, David Hows, Peter Membrey, Eelco Plugge, Apress, ISBN-13 (electronic): ISBN:978-1-4842-0895-3 6. Fullstack React: The Complete Guide to ReactJS and Friends, Anthony Accomazzo, Lean Publishing, Ari Learner, Clay Allsopp, David Guttman, Tyler McGinnis, Nate Murray, 7. The Road to React: Your journey to master React.js in JavaScript, by Robin Wieruch 8. Beginning React Native with Hooks, Greg Lim 9. Full-Stack React Projects: Learn MERN stack development by building modern web apps using MongoDB, Express, React, and Node.js, 2nd Edition 10. Angular From Theory To Practice, Asim Hussain, Version 1.2.0, 2017-11-24 11. Angular: Up and Running: Learning Angular, Step by Step, Shyam Seshadri, O'Reilly Media, Inc. 12. Mastering Web Application Development with AngularJS, Pawel Kozlowski Peter and Bacon Darwin, Packt Publishing
Teaching Methodology	Class Work, Discussion, Lab work, Self-Study, Seminars and/or Assignments, Practical implementation, Application development, Project development